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6. Construct a C program to implement pre-emptive priority scheduling algorithm.

Code:

main.c

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```c
#include<stdio.h>
int main(){
    int bt[10],rt[10],pr[10];
    int wt[10],tat[10];
    int n,i,time=0,smallest,complete=0;
    printf("Enter number of processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("Enter Burst Time: ");
        scanf("%d",&bt[i]);
        printf("Enter Priority: ");
        scanf("%d",&pr[i]);
        rt[i]=bt[i];
    }
    while(complete!=n)
    {
        smallest=-1;
        for(i=0;i<n;i++)
        {
            if(rt[i]>0 && (smallest==-1 || pr[i]<pr[smallest]))
                smallest=i;
        }
        rt[smallest]--;
        time++;
        if(rt[smallest]==0)
        {
            complete++;
            tat[smallest]=time;
            wt[smallest]=tat[smallest]-bt[smallest];
        }
    }
    printf("\nProcess\tWT\tTAT\n");
    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\n",i+1,wt[i],tat[i]);

    return 0;
}
```

Output:

Output

```
Enter number of processes: 3
Enter Burst Time: 5
Enter Priority: 2
Enter Burst Time: 3
Enter Priority: 1
Enter Burst Time: 4
Enter Priority: 3

Process WT  TAT
P1   3   8
P2   0   3
P3   8   12


=== Code Execution Successful ===
```